

Q6. Describe aetiology, stages , clinical features and management of Tuberculous Lymphadenitis(scrofula).

- **Definition:** Chronic **granulomatous inflammation** with **caseation necrosis** of the lymph node caused by members of the **M. tuberculosis** complex (e.g., M. tuberculosis, M. bovis).
- **Extrapulmonary TB:** Tuberculous lymphadenitis is one of the most common forms of extrapulmonary tuberculosis.

Morphology

- **Granuloma:** Characteristic structure made of:
 - **Langhans cells** (giant multinucleated cells).
 - **Epithelioid cell aggregates.**
 - **T cell lymphocytes** and few **fibroblasts.**
- **Caseous necrosis:** Granulomas evolve into necrotic tissue, often replacing lymphoid tissue.

Common Site

- Most commonly affects the **neck**, especially near the **sternocleidomastoid muscle.**
- Typically **unilateral, painless** initially.

Stages of Tuberculous Lymphadenitis

- **Stage 1:**
 - **Lymphoid hyperplasia:** Formation of tubercles and granulomas without caseation.
 - **Clinical:** Hard, bean-sized nodes; movable; normal skin without tenderness.
- **Stage 2 & 3:**
 - **Caseation necrosis:** Nodes enlarge, fuse, and become immovable.
 - **Clinical:** Skin becomes dark red, slightly hot, with fluctuating sensation. Possible **abscess formation.**
- **Stage 4:**
 - **Rupture:** Caseous material ruptures into surrounding tissue, forming a **confluent abscess.**
 - **Clinical:** Formation of **fistulae**, discharging pus with flocculent material. Skin becomes **dark purple.**

Clinical Features

- **Common:** Persistent, **painless swelling** of lymph nodes.
- **Progression:** Swollen nodes may discharge fluid.
- **Systemic symptoms:**
 - **Fever**
 - **Weight loss**
 - **Fatigue**
 - **Night sweats**

Diagnosis

- **Positive tuberculin test** (PPD test).
- **Chest X-ray** to check for pulmonary TB.
- **CT scan** for detailed lymph node assessment.
- **Cytology/biopsy** (e.g., FNAC).
- **AFB staining** to detect **acid-fast bacilli**.
- **Mycobacterial culture** for confirmation.

Treatment

- **Chemotherapy:**
 - **HRZE** for 2 months, followed by **HR** for 4 months.
- **DOTS:** Standard regimen for tuberculosis.
- **Surgery:** Indicated in cases of:
 - Persistent **fluctuant lesions**.
 - **Failure with chemotherapy**.
 - **Symptomatic relief**.

Q8. Cystic Hygroma (CH)

Definition

- The most common type of **lymphangioma**.
- **Congenital** multiloculated cystic lymphatic lesion filled with watery fluid.
- Can occur in any anatomical region.

Common Locations

- **Head and Neck (75% cases)**
 - Left-sided predilection.
 - Frequently affects the **posterior triangle of the neck**.
- **Axilla (20% cases)**

Microscopic Features

- Multiple **locules** filled with lymphatic fluid.
- Locules are **larger in deeper regions** and become smaller near the surface.
- Commonly forms in **loose areolar tissue**.

Clinical Features

- **Appearance**
 - Soft, **painless**, and **compressible (doughy)** masses.
 - Lesions typically **transilluminate** when light is passed through them.
- **Nature**
 - Benign but often **disfiguring** due to size and location.

Associated Complications

- May cause **airway obstruction** if large in size, especially in the neck.
- Risk of **infection** if the lesion ruptures.
- Difficulty in **surgical removal** due to proximity to vital structures.

Diagnosis

- **Ultrasound**: Used for initial assessment, showing cystic structures.
- **MRI/CT Scan**: Provides detailed localization and extent of the lesion.
- **Fine Needle Aspiration**: May help differentiate it from other lesions.

Treatment

- **Medical Management**
 - **Sclerosing agents** (e.g., bleomycin) can be used to shrink the lesion.
 - Regular follow-up for cases managed conservatively.
- **Surgical Treatment**
 - **Surgical excision** is the treatment of choice for complete removal.
 - Requires careful dissection to avoid damage to surrounding structures.
- **Advanced Options**
 - Laser therapy or radiotherapy in refractory cases.

Prognosis

- **Good prognosis** after complete surgical removal.
- Recurrence can occur if excision is incomplete.

Q9. Carotid Body Tumour (CBT)

- Also known as **chemodectoma** or **paraganglioma**.
- Develops within the **adventitia** of the medial aspect of the **carotid bifurcation**.

Types of Carotid Body Tumours

1. **Familial**
2. **Sporadic** (most common)
3. **Hyperplastic**

Aetiology

- **Known risk factors** include:
 - **Chronic hypoxic stimulation** (low oxygen levels over time).
 - **Genetic predisposition** (family history of similar tumours).

Clinical Features

- **Age of onset:** Most commonly around **45 years**.
- **Presentation:**
 - **Asymptomatic** mass in the **anterior triangle of the neck** (often the first sign).
 - Slow-growing, often **asymptomatic** for many years.
- **On examination:**
 - Mass is typically **vertically fixed** due to attachment to the **carotid bifurcation** (Fontaine sign).
 - **Bruit** (abnormal sound) may be heard when palpating the tumour.
 - **10%** of cases may present with **cranial nerve palsy** affecting:
 - **Hypoglossal nerve** (paralysis of the tongue).
 - **Glossopharyngeal nerve** (affecting swallowing and taste).
 - **Recurrent laryngeal nerve** (hoarseness).
 - **Spinal accessory nerve** (shoulder drop).
 - **Sympathetic chain** involvement (Horner syndrome: ptosis, miosis, anhidrosis).

Symptoms Associated with CBTs

- **Pain**
- **Hoarseness**
- **Dysphagia** (difficulty swallowing)
- **Horner syndrome** (drooping eyelid, constricted pupil)

- **Shoulder drop**

Treatment

- **Surgery** is the primary treatment.
- **Radiotherapy** may also be considered for some cases.

Q2. Recurrent Laryngeal Nerve Injury

Recurrent Laryngeal Nerve (RLN) Injury

Definition:

Damage to the recurrent laryngeal nerve (RLN), which controls the **vocal cords** and sensations in the **larynx**.

Causes:

1. **Thyroid surgery** – Most common cause.
2. **Neck surgery** – Includes carotid endarterectomy, esophageal surgery.
3. **Trauma** – Neck injury, including penetrating wounds.
4. **Tumors** – Thyroid, lung, or esophageal cancers.
5. **Infections** – Inflammation near the nerve.
6. **Intubation** – Misplacement of the endotracheal tube.

Symptoms:

- **Hoarseness** – Common in vocal cord paralysis.
- **Weak voice or loss of voice** – Especially with bilateral injury.
- **Breathing difficulty** – Particularly with both nerves affected.
- **Stridor** – Wheezing or high-pitched breathing.
- **Dysphagia** – Difficulty swallowing.
- **Aspiration** – Food or liquids entering the airway.

Diagnosis:

- **Laryngoscopy** – Checks vocal cord movement.
- **Electromyography (EMG)** – Assesses nerve function.
- **CT/MRI** – Identifies tumors or structural causes.

Management:

- **Observation** – Spontaneous recovery in some cases.
- **Voice therapy** – Improves voice quality and strength.
- **Surgical interventions:**

- **Thyroplasty** – To strengthen vocal cords.
- **Vocal cord injection** – Improves vocal cord positioning.
- **Reinnervation surgery** – For bilateral injuries.

Prevention:

- **Careful surgical technique** to avoid nerve damage.
- **Intraoperative nerve monitoring** to identify RLN during surgery.
- **Proper intubation** to prevent trauma to the RLN.

Complications:

- **Permanent hoarseness** or weak voice.
- **Aspiration pneumonia** due to chronic aspiration.
- **Airway obstruction** requiring emergency tracheostomy.
- **Psychological effects** like loss of confidence or depression.

Q3. Mention indications for Tracheostomy. Describe the steps of operation.

Definition: An operative procedure to create a surgical airway in the **cervical trachea**.

Indications:

1. **Bypass obstruction:**
 - **Congenital anomalies** (e.g., laryngeal hypoplasia, vascular web).
 - **Foreign body** that cannot be dislodged with Heimlich or BCLS.
 - **Supraglottic/glottic conditions** (e.g., infection, neoplasm, bilateral vocal cord paralysis).
2. **Neck trauma** causing injury to **thyroid/cricoid cartilages, hyoid bone, or great vessels**.
3. **Subcutaneous emphysema**.
4. **Facial fractures** leading to airway obstruction (e.g., comminuted midface and mandible fractures).
5. **Oedema** due to:
 - **Trauma, burns, infection, anaphylaxis**.
6. **Long-term mechanical ventilation** in **respiratory failure**.
7. **Pulmonary toilet:** Inadequate cough, **aspiration**, or difficulty handling secretions.
8. **Prophylaxis** for extensive head and neck procedures or convalescence.
9. **Severe sleep apnoea** not amendable to CPAP or other surgery.

Tracheostomy Procedure:

- **Position:** Unconscious or anaesthetized patient **supine** with neck extended. For conscious patients, sitting or semi-recumbent position.

- **Landmarks:** Palpate **thyroid notch**, **sternal notch**, and **cricoid cartilage**. Mark them.
- **Incision:**
 - **Vertical** 3-cm incision starting from cricoid cartilage.
 - Horizontal incisions can also be used for **better cosmetic results** but may trap secretions.
- **Dissection:**
 - Cut through **platysma** to expose midline **strap muscles**.
 - Retract **strap muscles** laterally.
 - Identify **pretracheal fascia** and **thyroid isthmus**, then divide the isthmus.
 - **Assess** proximity of **innominate artery** and control bleeding by cauterizing **jugular veins** and smaller vessels.

Tracheal Opening:

- **T-shaped:** 2-cm horizontal incision between **second and third** or **third and fourth** tracheal rings.
- **U- or H-shaped:** Tracheal flaps are reflected, beneficial in **obese patients**.
- **Permanent stoma:** Skin flaps sutured to a **rectangular tracheal opening** for long-term access.

Procedure Completion:

- **Enter trachea**, suction out **blood** and **secretions**.
- **Withdraw endotracheal tube** and insert the tracheostomy tube.
- Confirm airway integrity with **carbon dioxide return** and **bilateral breath sounds**.
- Secure the tracheostomy tube with **4-0 permanent sutures**.
- Attach a **tracheostomy collar** with the head flexed to prevent slack.
- **Avoid subcutaneous emphysema** by leaving the skin **open** and placing **iodine/petrolatum gauze** for 24 hours to prevent infection and manage minor oozing.

Q4. Discuss Postoperative Management and complications of Tracheostomy.

Postoperative Management :

- **Postoperative care** is critical. The trachea produces **copious secretions**, and regular **irrigation** with saline and **suctioning every 15 minutes** may be needed initially.
- **Suctioning** should be limited to **15 seconds** to avoid **tracheal ulceration** and **tracheitis**. Limit suctioning to the length of the tube to prevent airway injury.
- **Humidified oxygen** helps prevent **thickening of secretions**.
- **Mucolytic agents** like **acetylcysteine** (mucomyst) or **guaifenesin** can help loosen secretions. **Blocked cannula** can cause **life-threatening obstruction**.

- The **original tube** is left in place for **5-7 days** to allow healing. Afterward, sutures are removed, and the tube is **replaced**.
- For patients with acute tracheostomies, this is a chance to **downsize the tube** or change to a **metal (Jackson) tube**.
- Keep the site **clean and dry** to prevent infection from colonized bacteria.
- **Patient and family education** should begin as soon as possible.

Complications of Tracheostomy

A) Immediate Complications:

- **Apnoea** due to loss of **hypoxic respiratory drive**.
- **Bleeding**: From the vascular thyroid gland and lacerated vessels; must be **cauterized** or **ligated**.
- **Pneumothorax** or **pneumomediastinum**: Can occur due to injury to the **pleura** or **lung cupola** (especially in children).
- **Injury to adjacent structures**: Includes **recurrent laryngeal nerves**, **great vessels**, and **oesophagus**.
- **Postobstructive pulmonary oedema**.
- **Endotracheal tube ignition**: Rare, linked to **electrocautery** or **laser** use.

B) Early Complications:

- **Early bleeding**: Caused by increased blood pressure as the patient wakes up from anaesthesia and starts coughing.
- **Plugging with mucus**.
- **Tracheitis** (inflammation of the trachea).
- **Cellulitis** (infection around the site).
- **Displacement** of the tube.
- **Subcutaneous emphysema**: Caused by tight closure or false passage of the tube, can lead to **pneumothorax** or **pneumomediastinum**. Treatment includes loosening closure and performing a **tube thoracotomy**.
- **Atelectasis**: Can occur if the tube is too long, mimicking **unilateral mainstem intubation** and causing lung collapse.

C) Late Complications:

- **Bleeding**.
- **Tracheomalacia**: Weakening of the tracheal walls.
- **Tracheo-oesophageal fistula**: Abnormal connection between trachea and oesophagus.
- **Tracheocutaneous fistula**: Abnormal opening between trachea and skin.
- **Granulation tissue**: Excess tissue growth at the site.

Q6. Quinsy (Peritonsillar Abscess)

A **complication of tonsillitis**, quinsy is a **collection of pus** beside the tonsil in the **peritonsillar space**. This space is connected to deeper spaces like the **parapharyngeal** and **retropharyngeal** spaces, which can also become infected.

Quinsy often develops from **tonsillitis** → **cellulitis** → **abscess formation**.

Clinical Features

History

- Symptoms typically start **3-5 days** before evaluation.
- **Fever**
- **Malaise**
- **Headache**
- **Neck pain**
- **Severe throat pain** (usually on the affected side, may radiate to the ipsilateral ear)
- **Dysphagia** (difficulty swallowing)
- Change in **voice**
- **Otalgia** (ear pain)
- **Odynophagia** (painful swallowing)

Physical Evaluation

- **Mild/moderate distress**
- **Fever**
- **Tachycardia** (fast heart rate)
- **Dehydration**
- **Drooling**, trouble handling oral secretions
- **Trismus** (pain and spasm of jaw muscles)
- **Hot potato/muffled voice**
- **Rancid or fetid breath**
- **Cervical lymphadenitis** (swollen lymph nodes) in the anterior chain
- **Asymmetric tonsillar hypertrophy** (enlarged tonsil)
- **Localized fluctuance** (swelling of the abscess)
- **Inferior and medial displacement** of the tonsil
- **Contralateral deviation of the uvula** (uvula shifted to the opposite side)
- **Erythema** (redness) and **exudates** (pus) on the tonsil

Treatment

- **Needle aspiration** to drain the abscess and relieve pain.
- **Incision and drainage** may be required for larger abscesses.
- **Empiric antibiotics**, usually **clindamycin**, are recommended.

- **Antipyretics** (fever reducers) and **analgesics** (pain relievers) for symptom relief.
- If **airway is compromised**, **endotracheal intubation** may be necessary.
 - If intubation fails, a **cricothyroidotomy** or **tracheotomy** may be required.

Q7. Collar Stud Abscess

A **collar stud abscess** is an **acute suppurative infection** (pus formation) that typically affects a **digit** (finger), presenting as a **stud-like blister** on the surface.

Pathophysiology

- The abscess begins as a superficial blister and then **tracks to deeper tissue**.
- A **cold abscess** can form when the infection **tracks** through the **neurovascular bundle**, which are the nerves and blood vessels that pierce the **deep fascia** (the layer of connective tissue beneath the skin) and enter the **subcutaneous plane** (beneath the skin), forming a more extensive collection of pus.
- This deeper infection needs careful treatment, as the pus can spread and involve more structures in the finger.

Management

- **Simple incision of the blister** will not solve the problem, as it only addresses the superficial part of the infection.
- The finger must be thoroughly **explored** to identify the deeper abscess collection.
- Once the abscess is located, **delayed closure** of the wound is necessary to allow for drainage and healing. This approach helps avoid trapping pus in the tissue.

Treatment

- **Systemic antibiotics** are recommended to treat the infection and prevent its spread.
- The choice of antibiotic depends on the suspected or confirmed pathogen, and it's important to start antibiotics as soon as possible to avoid further complications.

Note: Prompt and appropriate intervention is crucial to prevent complications like **necrosis**, **sepsis**, or permanent damage to the finger's function.

Q8. Anatomy of thyroid gland

Anatomy of Thyroid Gland

- **Location:**
 - Lies in the **anterior neck**.
 - Extends from **C5 to T1 vertebral level**.

- Located **below the thyroid cartilage** and in front of the **trachea**.
- **Shape:**
 - **Butterfly-shaped** gland.
 - Has **two lobes** (right and left) connected by an **isthmus**.
 - Sometimes a **pyramidal lobe** is present (remnant of thyroglossal duct).
- **Coverings:**
 - Enclosed by a **fibrous capsule**.
 - Surrounded by **pretracheal fascia**, forming a sheath.
- **Relations:**
 - **Anterior:** Strap muscles (sternohyoid, sternothyroid, omohyoid).
 - **Posterior:** Trachea, esophagus, recurrent laryngeal nerve.
 - **Lateral:** Common carotid artery, internal jugular vein.
- **Blood Supply:**
 - **Arterial:**
 - Superior thyroid artery (branch of external carotid artery).
 - Inferior thyroid artery (branch of thyrocervical trunk).
 - **Venous:**
 - Superior and middle thyroid veins (drain into internal jugular vein).
 - Inferior thyroid vein (drains into brachiocephalic vein).
- **Nerve Supply:**
 - **Sympathetic:** From **cervical sympathetic ganglia**.
 - **Parasympathetic:** From **vagus nerve** (recurrent laryngeal nerve).
- **Lymphatic Drainage:**
 - Drains into **deep cervical** and **paratracheal lymph nodes**.
- **Functions:**
 - Produces **thyroid hormones (T3 and T4)** that regulate metabolism.
 - Secretes **calcitonin** (regulates calcium levels).
- **Clinical Importance:**
 - **Goiter:** Enlargement of the gland.
 - **Thyroidectomy:** Surgical removal of the gland.
 - **Thyroid nodules:** Benign or malignant growths in the gland.

Q9. Peripheral nerve injury

Peripheral Nerve Injury

- **Definition:**
Damage to peripheral nerves due to trauma, compression, or disease.
- **Causes:**
 - **Trauma** (cut, crush, stretch injury).
 - **Compression** (tumor, hematoma, tight cast).
 - **Iatrogenic** (surgical injury).
 - **Infections** (leprosy).
 - **Metabolic disorders** (diabetes, alcoholism).

- **Types of Nerve Injuries:**
 - **Neurapraxia:** Temporary conduction block without structural damage.
 - **Axonotmesis:** Axonal disruption with intact nerve sheath.
 - **Neurotmesis:** Complete nerve and sheath disruption.
- **Clinical Features:**
 - **Motor Symptoms:** Weakness or paralysis.
 - **Sensory Symptoms:** Numbness, tingling, or loss of sensation.
 - **Autonomic Symptoms:** Loss of sweating or temperature regulation.
- **Diagnosis:**
 - **Clinical Examination:** Muscle strength, reflexes, sensation.
 - **Electromyography (EMG):** Assesses nerve and muscle function.
 - **Nerve Conduction Studies (NCS):** Measures speed of nerve signals.
 - **Imaging:** MRI or ultrasound to detect compression or damage.
- **Management:**
 - **Conservative:** Rest, physiotherapy, and pain management.
 - **Surgical:** Nerve repair, grafting, or decompression.
 - **Rehabilitation:** Muscle re-education and occupational therapy.
- **Complications:**
 - **Chronic pain** (neuropathic pain).
 - **Permanent paralysis** or sensory loss.
 - **Contractures** due to muscle imbalance.
- **Prognosis:**
Depends on the **type**, **extent**, and **timing of intervention**.

Q2. What is mixed parotid tumor? Discuss its clinical features and management.

Pleomorphic Adenoma (Mixed Parotid Tumour)

- **Benign parotid tumour**, derived from **ductal (epithelial)** and **myoepithelial cells**.
- Shows **both epithelial and mesenchymal differentiation**.
- Named due to **architectural pleomorphism** (variable appearance under light microscopy).

Components

- **Epithelial cell component**
- **Myoepithelial cell component**
- **Stromal (mesenchymal) component**

Common Sites

- **Tail of superficial lobe of parotid gland (70-80%)**
- **Submandibular gland (10%)**
- Rare in **sublingual gland**

Key Points

- Though **benign**, it can grow large and undergo **malignant transformation** (forming **carcinoma ex-pleomorphic adenoma**).
- Tumour is **aneuploid**, can recur after resection, invade adjacent tissue, and rarely cause distant metastases.

Clinical Features

- **Solitary, slow-growing, firm, mobile, painless, nodular mass.**
- Colour varies: **whitish-tan to grey to bluish.**
- Size: Few mm to **large/giant** tumours.
- **Mobile**, unless located in the **palate.**
- In **parotid tail**, presents as **ear lobe eversion.**
- Rarely causes **facial nerve weakness**, unless large or malignant.
- In **deep lobe of parotid**: Presents as **oral retrotonsillar mass/parapharyngeal space tumour.**
- **Minor salivary gland tumours**: May cause **dysphagia, dyspnoea, hoarseness, chewing difficulty, epistaxis.**

Diagnosis

- **Tissue sampling procedures**
 - **FNA**: Sensitivity ~90%.
 - **Core needle biopsy**: Sensitivity ~97%.
 - Recommended before surgery.
- **Diagnostic imaging techniques**
 - **Ultrasound**: Characterizes superficial tumours.
 - **CT scan**: Visualizes tumour dimensions, **bony invasion.**
 - **MRI**: Superior for **soft tissue and perineural invasion.**

Treatment

- **Surgical resection** is the main treatment.
- For **parotid gland**:
 - **Superficial/total parotidectomy** (total preferred due to high recurrence).
 - Preserve **facial nerve** whenever possible.
- For **submandibular gland**:
 - **Simple excision** with preservation of **mandibular branch of trigeminal nerve, hypoglossal nerve, lingual nerve.**

Q3. What are the causes of acute Parotitis? Describe clinical features and treatment.

- **Inflammation of parotid glands.**

Aetiology of Acute Parotitis

1. **Acute bacterial parotitis**
2. **Acute viral parotitis (mumps)**

Acute Bacterial Parotitis

- Rare due to **antibiotics** and **IV fluids**.
- Previously common in **postoperative** or severely ill patients.
- Caused by **Staphylococcus aureus** or other **commensal bacteria**.

Acute Viral Parotitis (Mumps)

- Classic **childhood infection**, spread by **droplets** or direct contact with infected secretions (**paramyxovirus**).
- **MMR vaccine**:
 - 1st dose at **1 year**.
 - 2nd dose at **4-6 years**.
- Rare in vaccinated populations, but outbreaks may occur in unvaccinated teenagers or young adults.
- Characterized by:
 - **Enlarged, tender parotid glands**.
 - Pain during **parotid stimulation** (e.g., chewing).
- Complications: **Meningoencephalitis, pancreatitis, orchitis, deafness** (in young adults).

Clinical Features

- **Acute bacterial parotitis**:
 - **Progressive painful swelling, fever**, pain worsens with **chewing**.
 - **Purulent saliva** expressed upon gland massage.
- **Acute viral parotitis (mumps)**:
 - **Pain and swelling** lasting **5-9 days**.
 - **Bilateral involvement, moderate malaise, anorexia, fever**.
 - Gland tender on palpation.
 - **Clear saliva** expressed in normal glands.

Treatment

- **Acute bacterial parotitis**:
 - Caused by bacteria ascending from the mouth, common in **chronically ill** patients.
 - Severe cases caused by **methicillin-resistant Staphylococcus aureus (MRSA)**.
 - Investigations: **Gram stain, culture and sensitivity tests**.

- Empirical therapy: **Vancomycin 500 mg IV every 6 hours.**
- Modify treatment based on **culture results** or **infectious disease consultation.**

Q4. Describe pathology ,clinical features and management of Submandibular Salivary Calculus (Sialolithiasis).

- Most common in **submandibular gland.**
- **Frequent cause** of recurrent **sialadenitis.**

Reasons for Higher Frequency of Calculi Formation in Submandibular Gland

- **pH of saliva:** Alkaline in submandibular gland.
- **Viscosity of saliva:** More **mucous** in submandibular gland.
- **Anatomical factors:** Uphill course of **Wharton's duct.**

Common Location

- Near the **orifice of Wharton's duct** or at the **bend of the duct** behind the **mylohyoid muscle.**

Pathology

- Submandibular gland has **higher concentration of mucus**, making saliva more **viscous** compared to other glands.
- **Increased viscosity** and slower flow lead to **stasis** and **calculus formation.**

Clinical Features

- **Palpable lump** or **visible swelling** in the area of the gland.
- Pain, if present, originates from the **floor of the mouth.**
- **Intermittent swelling**, often **increased during meals.**
- **Complications:** Persistent obstruction leads to **bacterial infection** and **sialoadenitis.**

Treatment

1. **Small stones:**
 - **Hydration, moist heat, NSAIDs.**
 - Sucking on **bitter** or **sour** foods (e.g., **citrus fruits** like lemon/orange) may promote **salivation** and help expel the stone.
 - **Massage** by a specialist may help in some cases.
2. **For larger stones:**
 - **Duct cannulation** and **stone removal.**
 - **Gland excision** in recurrent cases.

- **Sialendoscopy.**
3. **Antibiotics:** For bacterial infection.

Q5. Mucous Cyst (Mucocoele) of the Oral Mucosa

Refers to **two related phenomena:**

- **Mucus extravasation phenomenon:** Swelling of connective tissue containing mucin due to a ruptured salivary gland duct, often caused by **local trauma**.
- **Mucus retention cyst:** Mucin collection due to an obstructed or ruptured salivary duct.

Clinical Features

- **Asymptomatic swellings** that **rapidly** appear and fluctuate in size.
- **Common location:** **Lower lip** (most common), also seen on the **buccal mucosa**, **anterior ventral tongue**, and **floor of the mouth (ranula)**.
- Superficial lesions appear **bluish** or **translucent**, while **deep lesions** have normal mucosal color.
- Size varies from **1 mm** to several **centimetres**. More common in **children and young adults**.
- Lesions may **enlarge rapidly** and then seem to **involute** due to rupture of contents into the oral cavity or resorption of mucus.
- History of **recent trauma** or **lip biting** may be reported, though many cases have no identified cause.
- Lesion duration: Typically **3-6 weeks**, but can range from **days** to **years** in rare cases.
- **Superficial mucocoeles** often appear as small fluid-filled vesicles on the **soft palate**, **retromolar pad**, **posterior buccal mucosa**, and occasionally the **lower labial mucosa**.
- **Granulation tissue** surrounds mucin, with **neutrophils** and **foamy histiocytes** present due to inflammation.
- **Intact mucosal lining**, but **repeated sucking** may lead to a **white, rough, keratotic surface**. Sometimes a **punctate sinus tract** may be seen, expressing **mucoid material**.
- Mucocoeles may **rupture spontaneously**, leaving an **ulcerated mucosal surface** that heals in a few days.
- **Chronic and recurrent** cases are common, with a history of **lichen planus**, **lichenoid drug reactions**, or **chronic graft-versus-host disease** involving the oral mucosa.

Treatment

- Some mucocoeles **resolve spontaneously** after a short time.
- Chronic mucocoeles may require **surgical removal**.

- **Recurrence** is possible, so the adjacent **salivary gland** may be excised as a preventive measure.
- Surgical options include **laser** and **minimally invasive techniques**, offering reduced recovery times.

Q6. Ranula

A **pseudocyst** associated with **mucus extravasation** into surrounding soft tissues.

- **Ranulas** are mucocoeles occurring in the **floor of the mouth** and usually involve the **major salivary glands**.
- Originates in the **sublingual gland**, **ducts of Rivini**, **Wharton's duct** of the **submandibular gland**, or rarely from the **minor salivary glands** at this location.

Types of Ranulas

1. **Oral ranulas**
 2. **Cervical (plunging) ranulas**
- **Oral ranulas:** Mucus extravasation pools **superior** to the **mylohyoid muscle**.
 - **Cervical ranulas:** Mucus extravasation occurs along the **facial planes** of the neck.

Clinical Features

- **Oral Ranula:**
 - **Asymptomatic swelling** in the **floor of the mouth**.
 - **Large, unilateral** blue to translucent mass resembling the belly of a frog.
 - **May cross the midline** when large, making it difficult to localize the salivary gland.
 - **Displaces the tongue** superiorly and medially, interfering with **speech, mastication, respiration, and swallowing**.
 - **Pressure on the lesion** may affect **submandibular salivary flow**, causing **pain** or discomfort when eating and **increased swelling** of the submandibular gland.
 - Consistency is **mucus-like**; lesion **does not blanch** on compression.
 - **Loss of bluish color** if located deeper in the floor of the mouth.
- **Cervical Ranula:**
 - **Asymptomatic**, continuously enlarging, **fluctuant** mass.
 - The **skin is usually intact**, and the mass is **freely movable** and **non-tender**.
 - Not associated with the **thyroid gland** or **lymph nodes**.
 - Mass may extend into the **mediastinum** and causes **lateral swelling**, occasionally crossing the midline.

Treatment

- **Oral Ranula:**

- **Surgical management** is preferred, including **complete excision** of the ranula and associated major salivary gland.
- **Laser ablation** and **cryosurgery**, either alone or after **marsupialization**, may be used.
- **Intracystic injections** of **OK-432** (streptococcal preparation) and **botulinum toxin A** have been reported as experimental treatments.
- **Cervical Ranula:**
 - Complete **surgical excision** of the oral portion and associated **sublingual salivary gland** (or rarely, the **submandibular gland**) is needed.

Q7. Frey's Syndrome

- **Food-related syndrome** that can be **congenital** or **acquired**, particularly after **parotid surgery**. It can persist for life.

Symptoms

- **Redness and sweating** on the **cheek area** near the ear, especially when eating, seeing, thinking about, or talking about certain foods that stimulate salivation.
- Sweating may occur after eating **lemon wedges**, which is often diagnostic for the condition.

Cause

- Results from **damage** to the **auriculotemporal branch** of the **trigeminal nerve** during **parotid gland surgery**.
- The **auriculotemporal nerve** normally carries **sympathetic fibres** to the sweat glands of the scalp and **parasympathetic fibres** to the parotid gland.
- After surgery, nerve fibres may regenerate **inappropriately**, causing the parasympathetic fibres (intended for saliva production) to stimulate **sweating** instead, leading to **gustatory sweating** (sweating in response to eating).

Treatment Options

- **Injection of botulinum toxin type A:** Temporarily blocks nerve signals to reduce sweating.
- **Surgical transection** of the nerve fibres: Cutting the nerve to stop abnormal sweating.
- **Application of anticholinergic ointment:** Ointments containing **scopolamine** can reduce sweating.

